- Costs
- Timing
- Scan Types. Sequential Scans, *Index Scans*, Index Only Scans
- Filtering. How many rows were filtered from a larger set?
- Indexes. Which indexes are named and were used? Are there indexes not listed that you expected to be there?
- Loops. For plan nodes, are loops greater than 1? This means the plan node was repeated.
- Ordering. Is sorting visible in the plan? Indexes are ordered; was an Index used for the sort operation?

For any Index Scans, compare the list of fields in the query to the fields that are indexed. Can the fields being selected in the query be reduced? If the selected fields match the indexed fields exactly, a more optimal *Index Only Scan* can be used over an Index Scan.

The `BUFFERS` argument to `EXPLAIN` provides insight about the amount of disk data that was fetched. The amount of data fetched can negatively impact query execution speed. Is it possible to select less data, such as fewer rows by using a LIMIT clause?

Look for `TIMESTAMP` and `TIMESTAMPTZ` columns in the query and whether those fields are indexed for the table. When these timestamp fields are covered by a B-tree index they are stored in a chronological ordering that can be useful for `ORDER BY` clauses to reduce the sort cost.

As you begin reading PostgreSQL query execution plans you may want to use additional tools to help you understand them. Free and commercial analysis tools are available. A query plan visualizer is built in to the `PgAdmin`[7] open source PostgreSQL GUI client. A free tool is also available at `explain.depesz.com`.[8]

`PgMustard` is a commercial tool for query plan analysis. Capture query execution plans from your planner and then upload the plan text to the PgMustard web client to get deep insights from the execution plan. While the tool is commercial, PgMustard does provide a free public `EXPLAIN` Glossary[9] that helps those learning all of the terms and concepts the query planner uses.

You've now started to read query execution plans and have seen some of the information you can learn. Next you'll dive into some specific query performance problems.

7. https://www.pgadmin.org
8. https://explain.depesz.com
9. https://www.pgmustard.com/docs/explain